Feature or Penalty? A Placement Diagnostic for Graph-Risk Signals in Learned Portfolio Decisions

Tenghan Zhong
University of Southern California
Los Angeles, California, USA
tenghanz@usc.edu

Abstract

Graph-derived domain signals create a placement problem in learned decision systems: they may inform the predictor, or they may enter the downstream decision layer directly. We study this question in learned portfolio contracts, where a graph-risk score can be supplied as a neural feature (Feature-only contract) or additionally priced as an explicit penalty on risky exposure (Graph-penalty contract). To separate placement from asset-specific graph-score ordering, we introduce a within-state permutation diagnostic. The control permutes the graph-risk scores across assets within each decision state, preserving the state-level marginal distribution and the explicit penalty placement while breaking the asset-specific graph-risk assignment. On Fama-French industry portfolios and an external universe of exchange-traded funds (ETFs), explicit penalty placement reduces conditional value-at-risk at the 95% level (CVaR95), crowding, and co-crash risk relative to feature-only placement over ten seeds, with essentially matched effort. All variants use matched architecture, decision rule, benchmark, and feasibility constraints. The diagnostic separates two channels: explicit penalty placement, not graph ordering, accounts for the aggregate CVaR reduction; the permutation control attains lower CVaR than the true ordering in both universes, while the true ordering helps most clearly for industry crowding. We therefore identify explicit decision-layer placement as the more reliable source of risk control in this setting.

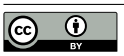
CCS Concepts

• **Information systems** → **Decision support systems**; *Data analytics*; • **Computing methodologies** → **Machine learning**; • **Applied computing** → *Economics*.

Keywords

graph-based decision systems, signal placement, permutation diagnostics, financial risk control, portfolio contracts, systemic risk

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1 Introduction

Graph-derived signals are increasingly used in learned decision systems, but standard feature-based use does not determine whether the signal should also enter the downstream decision rule directly. We study this question in delegated portfolio risk control, an instance of graph-based decision support where a graph-risk score can remain in the feature map or be additionally priced through an explicit exposure penalty in the decision layer. Financial graph signals model relations among traded assets [18, 25]; in risk-control settings, these relations often encode tail dependence, crowded trading, and systemic exposure in financial networks [1, 12, 19]. In portfolio-delegation contracts, a principal compensates an agent for output while limiting downside and systemic exposure [5, 16]. The key question is placement; the randomized-score comparison diagnoses whether the graph-derived ordering adds metric-specific value beyond that placement.

Much of financial graph learning concentrates on the signal itself, through richer graph estimators, attention mechanisms, or tail-aware encoders. The implicit assumption is that more informative signals propagate to better decisions. A flexible neural learner can in principle use such a signal through its feature map, but its marginal effect is then learned indirectly and can be entangled with other predictors, finite-sample noise, and the training objective. The same signal written into the contract changes the marginal price of risky exposure by construction. Standard feature-removal ablations cannot separate these two channels: they conflate the loss from absent information with the loss from absent placement, leaving open whether placement effects exist beyond signal informativeness.

Related work on decision-focused learning trains predictive components through downstream decisions [9, 20, 22, 24], while differentiable optimization layers embed optimization problems inside neural architectures [3, 6]. Algorithmic contract design instead studies how to compute globally optimal contracts under specified information structures [10, 14, 23]. Our goal is different: we fix the contract's functional form, learn its coefficients, and study where in the contract a graph-risk signal should enter. The contribution is therefore a placement–ordering diagnostic.

We instantiate the diagnostic in portfolio delegation, treating industry portfolios or exchange-traded funds (ETFs) as delegated assets. A learner maps asset features and a pre-contract graph-risk score to payment-rule coefficients; the agent's best-response exposure is computed by an analytic feasibility step that enforces incentive compatibility, individual rationality, and a minimum effort bound. The graph-penalty contract is first compared with the feature-only contract to measure explicit penalty placement, and

then with the randomized-score contract to test whether true graph-score ordering adds value under the same explicit penalty placement.

Our contributions are as follows:

- **Separable placement and ordering channels.** We frame the value of a graph-derived signal in a learned decision rule as two separately testable design questions: where the signal enters the decision rule, and how it ranks assets within a decision state. This framing exposes distinct failure modes: a graph score can be informative yet weakly placed, or explicitly placed but metric-specific in its ordering value.
- **Within-state permutation control.** We introduce a score-marginal-matched randomized control that preserves the within-state score distribution and explicit penalty placement while breaking asset-specific ordering. Unlike feature-removal ablations, this control tests graph-specific value under the same explicit penalty placement.
- **Two-universe empirical evidence.** We instantiate the diagnostic in Fama–French industry portfolios and an external ETF universe, showing that explicit penalty placement consistently reduces downside and systemic-risk metrics, while graph-specific value is metric-dependent. Effort checks and node-level exposure diagnostics support the intended exposure-reweighting channel.

2 A Within-State Permutation Diagnostic for Learned Portfolio Contracts

Graph-risk score. Let N denote the number of portfolio nodes and let t denote a decision state. For nodes $i, j \in \{1, \dots, N\}$, we compute a public rolling co-tail adjacency matrix $A_t \in \mathbb{R}^{N\times N}$ over a lookback window of length L. Its entry $A_{ij,t}$ is the fraction of lookback days on which nodes i and j both fall below their rolling 5% return quantiles. This co-tail construction is motivated by tail comovement and financial connectedness measures for systemic exposure [2, 7]. Let $\mathbf{1}_N$ be the N-vector of ones, let $D_t = \mathrm{diag}(A_t\mathbf{1}_N)$ be the tail-degree matrix, let I_N be the $N \times N$ identity matrix, let $\delta = 10^{-3}$ be an identity-shift stabilizer, and set $H_t = D_t + A_t + \delta I_N$. Let $\ell_t \in \mathbb{R}^N_+$ denote rolling lower-tail loss magnitudes and let $\varepsilon_{\mathrm{num}} > 0$ be a numerical stabilizer. The public graph-risk score of node i is

$$m^G_{i,t} = \frac{(H_t\ell_t)_i}{N^{-1}\sum_{j=1}^N (H_t\ell_t)_j + \varepsilon_{\mathrm{num}}}.$$

The score is pre-contract, nonnegative, and normalized around one, so $m^G_{i,t} > 1$ denotes above-average graph-weighted lower-tail exposure.

Contract variants. For a fixed decision state, write $m^G_i = m^G_{i,t}$. Let ξ_i denote the compensation paid to agent i, b_i the base transfer, X_i output, $\bar{X}^{\mathrm{GLS}}_i$ the covariance-regularized generalized least squares (GLS) peer benchmark [4], θ_i the own-output loading, η_i the peer-benchmark loading, λ_i the graph-penalty loading, and $\hat{s}_i$ realized risky exposure. Agent i's contract is

$$\xi_i = b_i + \theta_i X_i + \eta_i(X_i - \bar{X}^{\mathrm{GLS}}_i) - \lambda_i m^G_i \hat{s}_i.$$

The GLS benchmark is $\bar{X}^{\mathrm{GLS}}_i = \sum_{j\neq i} w^{\mathrm{GLS}}_{ij} X_j$, where the weights w^{GLS}_{ij} minimize $\mathrm{Var}(X_i - \sum_{j\neq i} w^{\mathrm{GLS}}_{ij} X_j)$ subject to $\sum_{j\neq i} w^{\mathrm{GLS}}_{ij} = 1$, with covariance regularization computed from the rolling lookback window. The peer-benchmark term follows relative-performance evaluation logic for filtering common shocks in incentive contracts [15, 17]. The feature-only variant observes m^G_i but sets $\lambda_i = 0$. The graph-penalty variant keeps the same features and allows $\lambda_i > 0$, giving $m^G_i\hat{s}_i$ an explicit marginal price. To isolate the explicit penalty channel, the constant-score control keeps m^G_i in the feature map but sets the explicit penalty multiplier to one, yielding $-\lambda_i\hat{s}_i$. This separates the graph-score multiplier from an unscored heterogeneous penalty channel. The randomized-score variant uses, for each seed, a fixed within-state permutation of node scores in both the feature map and the same explicit penalty term. This yields a node-level mechanism prediction: relative to the feature-only contract, explicit graph-risk pricing should contract deployed exposure more strongly for nodes with larger pre-contract m^G_i. For visual reference, the top panels of Figure 1 also include a no-graph neural baseline that omits m^G_i from the feature map and sets $\lambda_i = 0$. This baseline is not part of the formal diagnostic and is omitted from Table 1.

Learner and diagnostic contrasts. Given the state features, including m^G_i, the neural component outputs the effective incentive intensity z_i, the peer-benchmark loading η_i, and the graph-penalty loading λ_i; the own-output loading is then $\theta_i = z_i - \eta_i$. A constrained output step keeps incentives nonnegative and above the minimum effort bound. Given these coefficients, an analytic feasibility step computes each agent's best-response exposure s^*_i, distinct from the realized exposure $\hat{s}_i$ entering the contract formula, and sets the base payment b_i so that participation binds. Placement compares Graph-penalty contract with Feature-only contract, whereas ordering compares Graph-penalty contract with Randomized-score contract. Given the rolling tail indicators, constructing the co-tail graph costs $O(LN^2)$ time and $O(N^2)$ storage per state, while constructing the within-state permutation adds $O(N)$ time and storage.

3 Empirical Placement–Ordering Diagnostic

Data and evaluation protocol. We evaluate on two traded-asset universes. The main universe is the Fama-French 49-industry portfolio panel [13]; a universe of 20 ETFs, constructed from daily adjusted ETF prices from Yahoo Finance, provides an external validation setting [26]. The ETF ticker list and preprocessing rules are provided in the artifact. Each decision state uses a 252-day lookback and a 20-day horizon. Training uses 50 bootstrap scenario paths per decision state for computational efficiency. Reported test-set risk metrics are computed by re-evaluating the trained models with 500 common bootstrap scenario paths per state. Residual shocks $\epsilon \in \mathbb{R}^N$ are propagated as $\tilde{\epsilon} = (I_N - \rho W_t)^{-1}\epsilon$, where W_t is the row-normalized version of A_t and $\rho = 0.25$ is the propagation parameter. The industry split is chronological: train 1990–2005 (190 states), validation 2006–2007 (25 states), and test 2008–2026Q1 (228 states). The ETF split uses train 2015–2018 (28 states), validation 2019 (13 states), and test 2020–2026Q1 (77 states). Learned models are trained over ten fixed seeds.

The learner is a two-hidden-layer ReLU network of width 32, trained with Adam [21], with $z_i \in [0.80, 1.20]$ enforcing the effort-preserving range. All learned variants solve the feasibility system during training rather than as a post-hoc correction, so learned coefficients are computed under the same feasibility constraints that

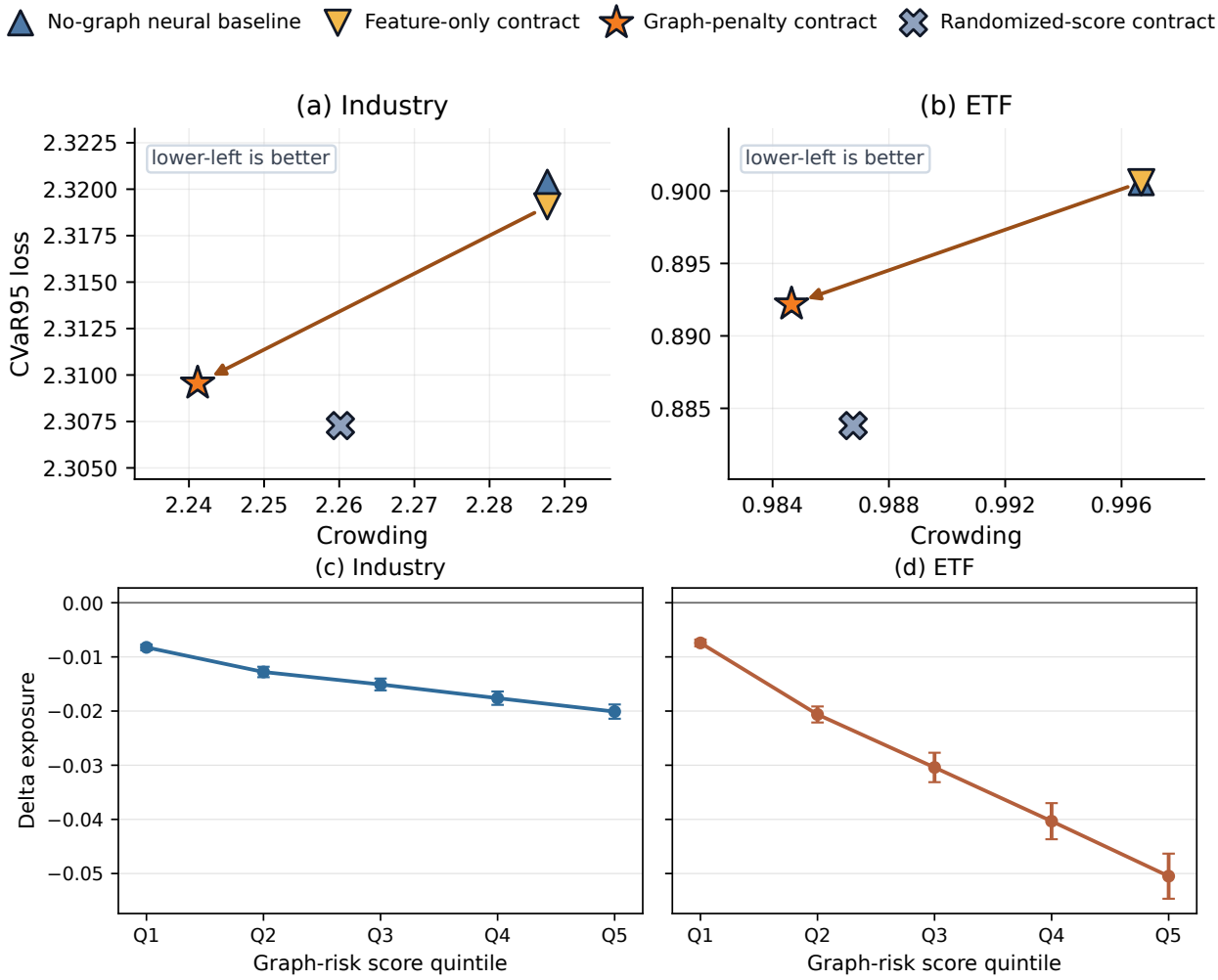


Figure 1: Aggregate and node-level diagnostics. Top: Crowding versus CVaR95 over ten seeds for the industry and ETF universes; lower-left is better. Randomized-score contract preserves explicit penalty placement while permuting graph-risk scores, testing ordering value. Bottom: On matched test states, we recompute each asset's optimal exposure s_i^* and report $\Delta s_i^* = s^*_{i,\text{GraphPenalty}} - s^*_{i,\text{FeatureOnly}}$ by graph-risk quintile. More negative values indicate stronger exposure contraction under explicit graph-risk pricing. Error bars show 95% intervals across seed-level quintile means.

govern evaluation. The three formal variants in Table 1 optimize the same risk-sensitive objective: mean principal payoff penalized by CVaR95 loss, payment variation, crowding, effort shortfall below 0.04, and coefficient smoothness. The corresponding weights are 0.65, 0.05, 0.10, 50, and 0.01. These weights are held identical across the tabled learned variants and were fixed before the reported test-set evaluation; the constant-score control uses the same training and evaluation protocol.

We report outcomes aligned with the diagnostic. CVaR95 is computed over all test states and 500 re-evaluation scenario paths and measures lower-tail downside exposure. Crowding[1] measures how concentrated the induced risk-taking is across nodes. Co-crash[2] measures joint lower-tail exposure. Principal certainty equivalent (CE) and raw payoff are secondary outcomes because the explicit penalty is designed to trade off upside for lower downside and systemic exposure.

All reported contrasts are paired on the same test states and scenario draws; intervals are 95% seed-clustered bootstrap intervals [8, 11], and tests aggregate across the ten training seeds.

Figure evidence. The top panels of Figure 1 give the aggregate diagnostic: Graph-penalty contract moves from Feature-only contract

[1]For decision state t, let $s_t^* = (s^*_{1,t}, \ldots, s^*_{N,t})^\top \in \mathbb{R}^N$ denote the induced risky-exposure vector, and let $Q_t = H_t = D_t + A_t + 10^{-3} I_N$ denote the evaluation tail-risk matrix. The crowding score is $c_t = (s_t^*)^\top Q_t s_t^* / N$.

[2]Co-crash is the fraction of bootstrap scenario paths in which at least 20% of nodes fall below their rolling 5% return quantiles.

Table 1: Combined placement–ordering diagnostic. Feature, Graph, and Rand. report mean risk outcomes for the Feature-only contract, Graph-penalty contract, and Randomized-score contract, respectively. Placement deltas are Graph-penalty contract minus Feature-only contract; ordering deltas are Graph-penalty contract minus Randomized-score contract. Lower is better for all reported metrics. CVaR95 uses pooled test-set payoff losses; crowding and co-crash are averaged over matched test states. CI denotes 95% confidence interval.

		Mean outcome			Paired delta [95% CI]	
Dataset	Metric	Feature	Graph	Rand.	Placement	Ordering
Industry	CVaR95 loss	2.3191	2.3096	2.3073	−0.0095 [−0.0109, −0.0079]	+0.0023 [+0.0009, +0.0039]
Industry	Crowding	2.2877	2.2412	2.2602	−0.0465 [−0.0510, −0.0424]	−0.0190 [−0.0224, −0.0159]
Industry	Co-crash	0.1060	0.1037	0.1046	−0.0023 [−0.0026, −0.0021]	−0.0008 [−0.0011, −0.0007]
ETF	CVaR95 loss	0.9006	0.8922	0.8838	−0.0084 [−0.0094, −0.0074]	+0.0084 [+0.0075, +0.0093]
ETF	Crowding	0.9967	0.9847	0.9868	−0.0120 [−0.0154, −0.0090]	−0.0021 [−0.0051, +0.0001]
ETF	Co-crash	0.1743	0.1733	0.1730	−0.0010 [−0.0014, −0.0007]	+0.0003 [−0.0001, +0.0006]

toward lower crowding and lower CVaR95 loss, while Randomized-score contract keeps the explicit penalty term but breaks the graph-score ordering; in both universes, Randomized-score contract attains lower CVaR than Graph-penalty contract. The bottom panels examine the allocation channel behind this movement: higher graph-risk quintiles show stronger exposure contraction under Graph-penalty contract relative to Feature-only contract. Together, the panels connect the contract term $-\lambda_i m_i^G \hat{s}_i$ to both asset-level exposure changes and aggregate downside-risk reductions.

Table 1 reports the placement–ordering diagnostic. The mean columns give the absolute risk levels of the feature-only, graph-penalty, and randomized-score contracts; the paired contrasts give the two diagnostic comparisons. Placement is the primary claim: Graph-penalty contract lowers CVaR95 loss, crowding, and co-crash risk relative to Feature-only contract in all six pre-specified universe–metric cells, with the 95% seed-clustered bootstrap intervals for these risk deltas excluding zero. The same sign appears for every seed in all six cells. Relative reductions are 0.4–2.2% in the industry universe and 0.6–1.2% in the ETF universe, incremental to a feature-only contract with GLS benchmarking, an effort floor, and participation adjustment. These improvements represent risk control rather than payoff dominance: relative to Feature-only contract, raw principal payoff under Graph-penalty contract is lower by about 0.0077 in the industry universe and 0.0017 in the ETF universe, while principal CE is mixed.

The constant-score control sharpens this attribution: it keeps m_i^G in the feature map but sets $m_i^G \equiv 1$ inside the explicit penalty; relative to Feature-only contract, it improves all six risk cells, with CVaR95 deltas of −0.0094 and −0.0163 in the industry and ETF universes, respectively, and all six CIs excluding zero. Relative to this control, Graph-penalty contract is not better on aggregate CVaR (industry −0.0002, CI spans zero; ETF +0.0079, CI excludes zero) but lowers crowding in both universes (industry −0.0079, ETF −0.0043; CIs exclude zero).

Graph-specific ordering value is secondary and metric-specific. Relative to Randomized-score contract, the true graph ordering raises CVaR95 loss in both universes, improves industry crowding and co-crash risk, and gives mixed ETF ordering effects. Thus the randomized-score and constant-score controls provide converging evidence: aggregate-CVaR control comes from the explicit exposure penalty.

The risk reduction is not driven by effort substitution: mean effort changes by -4.4×10^{-6} (industry) and 1.2×10^{-5} (ETF), with both intervals including zero, supporting the node-level reweighting channel in the bottom panels of Figure 1.

4 Conclusion

Whether a domain signal is informative and where it should enter a learned decision rule are separately testable design questions, and the within-state permutation diagnostic separates them. In portfolio-delegation contracts on Fama–French industries and ETFs, placement yields directionally consistent risk reductions across all six universe–metric cells, with the same sign across every seed. The explicit penalty channel, isolated by a constant-score control, accounts for the aggregate CVaR reduction, while the graph-score multiplier carries the crowding benefit, clearest in the industry universe. We view these as distinct design dimensions: progress on graph estimators and progress on signal placement should be evaluated separately, and a learner that already consumes a graph score may still benefit from explicit placement. A natural next step is to characterize when cross-sectional graph ordering survives the downstream best-response map and adds value beyond explicit placement, and to use that characterization to co-design dynamic or learned graph estimators with the decision layer.

Code and Artifact Availability. Code repository is available at https://github.com/TenghanZhong/within-state-score.

GenAI Usage Disclosure

During the preparation of this manuscript, generative AI tools were used only for limited writing and editing support, including grammar correction, sentence rephrasing, and language polishing. No generative AI tools were used to generate technical ideas, experimental results, analytical claims, or scientific conclusions. All methodological decisions, implementation, experiments, and final manuscript content were produced and verified by the author.

References

[1] Daron Acemoglu, Asuman Ozdaglar, and Alireza Tahbaz-Salehi. 2015. Systemic Risk and Stability in Financial Networks. *American Economic Review* 105, 2 (2015), 564–608. doi:10.1257/aer.20130456

[2] Tobias Adrian and Markus K. Brunnermeier. 2016. CoVaR. *American Economic Review* 106, 7 (2016), 1705–1741. doi:10.1257/aer.20120555

[3] Akshay Agrawal, Brandon Amos, Shane Barratt, Stephen Boyd, Steven Diamond, and J. Zico Kolter. 2019. Differentiable Convex Optimization Layers. In *Advances in Neural Information Processing Systems 32*. Curran Associates, Inc., 9562–9574. https://papers.neurips.cc/paper/9152-differentiable-convex-optimization-layers

[4] A. C. Aitken. 1936. On Least Squares and Linear Combination of Observations. *Proceedings of the Royal Society of Edinburgh* 55 (1936), 42–48. doi:10.1017/S0370164600014346

[5] Rui Albuquerque, Luís Cabral, and José Guedes. 2019. Incentive Pay and Systemic Risk. *The Review of Financial Studies* 32, 11 (2019), 4304–4342. doi:10.1093/rfs/hhz028

[6] Brandon Amos and J. Zico Kolter. 2017. OptNet: Differentiable Optimization as a Layer in Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 70)*. PMLR, 136–145. https://proceedings.mlr.press/v70/amos17a.html

[7] Monica Billio, Mila Getmansky, Andrew W. Lo, and Loriana Pelizzon. 2012. Econometric Measures of Connectedness and Systemic Risk in the Finance and Insurance Sectors. *Journal of Financial Economics* 104, 3 (2012), 535–559. doi:10.1016/j.jfineco.2011.12.010

[8] A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. 2008. Bootstrap-Based Improvements for Inference with Clustered Errors. *The Review of Economics and Statistics* 90, 3 (2008), 414–427. doi:10.1162/rest.90.3.414

[9] Priya L. Donti, Brandon Amos, and J. Zico Kolter. 2017. Task-Based End-to-End Model Learning in Stochastic Optimization. In *Advances in Neural Information Processing Systems 30*. Curran Associates, Inc., 5484–5494. https://papers.neurips.cc/paper/7132-task-based-end-to-end-model-learning-in-stochastic-optimization

[10] Paul Dütting, Tim Roughgarden, and Inbal Talgam-Cohen. 2019. Simple versus Optimal Contracts. In *Proceedings of the 2019 ACM Conference on Economics and Computation (EC '19)*. ACM, 369–387. doi:10.1145/3328526.3329591

[11] Bradley Efron and Robert J. Tibshirani. 1993. *An Introduction to the Bootstrap*. Chapman and Hall, New York.

[12] Larry Eisenberg and Thomas H. Noe. 2001. Systemic Risk in Financial Systems. *Management Science* 47, 2 (2001), 236–249. doi:10.1287/mnsc.47.2.236.9835

[13] Kenneth R. French. 2026. Kenneth R. French Data Library: 49 Industry Portfolios [Daily]. Accessed: 2026-05-21. https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_49_ind_port.html

[14] Tomer Galanti, Aarya Bookseller, and Korok Ray. 2025. Scalable Principal-Agent Contract Design via Gradient-Based Optimization. arXiv:2510.21177 [cs.LG] doi:10.48550/arXiv.2510.21177

[15] Robert Gibbons and Kevin J. Murphy. 1990. Relative Performance Evaluation for Chief Executive Officers. *Industrial and Labor Relations Review* 43, 3 (1990), 30S–51S. doi:10.1177/001979399004300303

[16] Bengt Holmström. 1979. Moral Hazard and Observability. *The Bell Journal of Economics* 10, 1 (1979), 74–91. doi:10.2307/3003320

[17] Bengt Holmström. 1982. Moral Hazard in Teams. *The Bell Journal of Economics* 13, 2 (1982), 324–340. doi:10.2307/3003457

[18] Yoonsik Hong and Diego Klabjan. 2025. Graph Learning for Foreign Exchange Rate Prediction and Statistical Arbitrage. In *Proceedings of the 6th ACM International Conference on AI in Finance*. Association for Computing Machinery, 692–699. doi:10.1145/3768292.3770418

[19] Amir E. Khandani and Andrew W. Lo. 2011. What Happened to the Quants in August 2007? Evidence from Factors and Transactions Data. *Journal of Financial Markets* 14, 1 (2011), 1–46. doi:10.1016/j.finmar.2010.07.005

[20] Juchan Kim, Inwoo Tae, and Yongjae Lee. 2025. Estimating Covariance for Global Minimum Variance Portfolio: A Decision-Focused Learning Approach. In *Proceedings of the 6th ACM International Conference on AI in Finance*. Association for Computing Machinery, 105–113. doi:10.1145/3768292.3770378

[21] Diederik P. Kingma and Jimmy Ba. 2015. Adam: A Method for Stochastic Optimization. In *International Conference on Learning Representations*. https://arxiv.org/abs/1412.6980

[22] Jayanta Mandi, James Kotary, Senne Berden, Maxime Mulamba, Victor Bucarey, Tias Guns, and Ferdinando Fioretto. 2024. Decision-Focused Learning: Foundations, State of the Art, Benchmark and Future Opportunities. *Journal of Artificial Intelligence Research* 80 (2024), 1623–1701. doi:10.1613/jair.1.15320

[23] Kanstantsin Pashkovich and Jacob Skitsko. 2026. Linear Contracts for Supermodular Functions Based on Graphs. In *Algorithmic Game Theory: 18th International Symposium, SAGT 2025, Bath, UK, September 2–5, 2025, Proceedings (Lecture Notes in Computer Science, Vol. 15953)*. Springer, 422–438. doi:10.1007/978-3-032-03639-1_23

[24] Bryan Wilder, Bistra Dilkina, and Milind Tambe. 2019. Melding the Data-Decisions Pipeline: Decision-Focused Learning for Combinatorial Optimization. *Proceedings of the AAAI Conference on Artificial Intelligence* 33, 01 (2019), 1658–1665. doi:10.1609/aaai.v33i01.33011658

[25] Sheng Xiang, Dawei Cheng, Chencheng Shang, Ying Zhang, and Yuqi Liang. 2022. Temporal and Heterogeneous Graph Neural Network for Financial Time Series Prediction. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*. Association for Computing Machinery, 3584–3593. doi:10.1145/3511808.3557089

[26] Yahoo Finance. 2026. Yahoo Finance Historical Market Data. Accessed: 2026-05-23. https://finance.yahoo.com/